JUAN CAMILO VASQUEZ CORREA

PhD Computer Science | AI researcher

SPEECH RECOGNITION | AI FOR HEALTHCARE | NATURAL LANGUAGE PROCESSING | MULTIMODAL AI

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PROFILE

AI Research Scientist (PhD) with 9+ years of interdisciplinary R&D experience in generative speech and language models, multilingual ASR, speech-driven avatar animation, and emotion recognition. Passionate about building human-centered, generative AI systems and advancing personalization in large-scale environments. Proven track record of bridging academic research and industrial innovation, from designing scalable AI solutions to publishing in top-tier venues (ICASSP, Interspeech, etc.). Skilled at leading projects across domains, collaborating with multidisciplinary teams, and translating cutting-edge AI/ML techniques into impactful applications for accessibility, healthcare, and human–computer interaction.

EXPERIENCE

07/2022 – present **Vicomtech Foundation (Spain)** — **Associate Researcher**

- Designed, trained, and deployed bilingual ASR systems (Spanish–Basque), achieving state-of-the-art performance and winning a competitive challenge (paper here)
- Designed and trained AI models for real-time speech-driven avatar animation using ARKit Blendshapes. (paper here)
- Built noise-robust ASR to improve accessibility for visually impaired users in Madrid Metro vending machines.
- Trained and deployed multilingual deepfake detection systems, improving robustness to unseen synthesis methods.

Technologies used: PYTORCH, TRANSFORMERS (HUGGING FACE), SCIKIT-LEARN, PANDAS, DOCKER, GIT, FASTAPI, ONNX

06/2021 – 07/2022 **Pratech Group (Colombia)** — **Head of R&D&i**

- Led R&D projects leveraging AI and natural language understanding to enhance customer service.
- Designed and led the implementation of software architecture for customer service chatbots.
- Defined and executed research & innovation strategies, aligning AI solutions with product goals.

Technologies used: PYTHON, DOCKER, IBM WATSON, AZURE, PYTORCH

04/2018 – 04/2021 **Pattern Recognition Lab, Friedrich Alexander University Erlangen-Nuremberg (Germany)** — **Staff Researcher**

- Conducted speech and NLP research for digital biomarkers in neurodegenerative disease assessment (see Here).
- Authored and published papers in top-tier conferences and journals (ICASSP, Interspeech).
- Supervision of Bachelor and Master thesis in Biomedical Engineering and Computer Science.

Technologies used: PYTHON, PYTORCH, TENSORFLOW, KALDI

2016 – 2020 **Department of Electronics and Telecommunications Engineering, University of Antioquia (Colombia)** — **Lecturer and Software Developer**

- Taught Digital Signal Processing at the undergraduate level, designing and delivering curriculum.(Content (in Spanish))
- Researched speech and biomedical signal processing for healthcare applications.
- Software Development to evaluate the speech production of patients with Parkinson's disease (Neurospeech).

2015 – 2016 **Telefónica R&D (Spain)** **Intern Researcher**

· Designed speech processing algorithms for noisy environments to extract paralinguistic traits.

EDUCATION

- **Doktor Ingenieur (Ph.D.)**, Dep. of Informatik, Friedrich Alexander Universität Erlangen-Nürnberg, Germany, 2022. *Multimodal Assessment of Parkinson's Disease Patients Using Information from Speech, Handwriting, and Gait.*
- **MSc in Telecommunications Engineering**, University of Antioquia, Colombia, 2016. *Emotion Recognition from Speech with Acoustic, Non-Linear and Wavelet-based Features Extracted in Different Acoustic Conditions.*
- **Bachelor in Electronics Engineering**, University of Antqioquia, Colombia, 2013.

LANGUAGES

Spanish C2 Proficient User | Mother tongue

English C1 Proficient User

PROUD OF

Open Source Contributions
Creator of **DisVoice**, a widely used Python library for speech feature extraction in healthcare and speech technology research.

Social contribution
Mentor in Misión TIC 2020, teaching programming fundamentals to youth and advancing digital inclusion in Colombia.

Awards and achievements
2024 - Winner, Bilingual Basque–Spanish ASR Challenge (IberSpeech), achieving state-of-the-art ASR performance.
2019 Best Paper Award, Iberoamerican Congress on Pattern Recognition (CIARP), for Paper: *Convolutional Neural Networks and a Transfer Learning Strategy to Classify Parkinson's Disease from Speech in Three Different Languages*
2019 First prize INTERSPEECH hackaton, designing of a skill for Alexa devices.
2017 IEEE Travel grant for the International Conference on Acoustics, Speech, and Signal Processing (ICASSP).
2015 International Speech and Communication Association Student travel grant to attend INTERSPEECH conference.
2014 Young Researcher and Innovator COLCIENCIAS (Colombian Ministry of Science).

R&D PROJECTS

- **ANIMEX**, Basque Government. 2024-2026. Animation of expressive avatars through audio and motion capture.
  Lead researcher on generative audio-driven avatar animation, combining speech synthesis, emotion recognition, and multimodal modeling for real-time and expressive avatars.
- **GRACE**, European Union, Horizon 2020-2023
  Global Response Against Child Exploitation.
  Developed multilingual speech recognition and keyword spotting technologies for law enforcement, handling large-scale multilingual datasets and adapting models to unseen domains.
- **APPRAISE**, European Union, Horizon 2020-2023
  Facilitating private and public security operators to mitigate terrorism scenarios against soft targets.
  Built speech recognition tools for threat intelligence in online content, focusing on scalable pipelines for multilingual data and robust performance in real-world conditions.
- **TAPAS**, European Union, Horizon 2018-2022
  Training Network on Automatic Processing of PAthological Speech. Research and innovation programme under Marie Sklodowska-Curie actions: European Union's Horizon 2020.
  Early-stage researcher in automatic processing of pathological speech, modeling disease progression with multimodal time-series (speech, handwriting, gait) and contributing to long-term personalization of digital health tools.

PUBLICATIONS

Author of **90+ peer-reviewed publications** in top international conferences and journals on **machine learning, speech processing, natural language processing, and multimodal AI**. Work published at venues such as **ICASSP, Interspeech**, and others.
Selected Publications:

- Vasquez-Correa, J.C., et al. (2025) *Emotion-Aware Speech-Driven Facial Avatar Animation via Joint Blendshape Prediction and Emotion Recognition*. Lecture Notes in Artificial Intelligence.
- Vasquez-Correa, J.C., et al. (2024). *The Vicomtech Speech Transcription Systems for the Albayzin 2024 Bilingual Basque-Spanish Speech to Text (BBS-S2T) Challenge*. IberSpeech.
- Vasquez-Correa, J.C., et al. (2018). *Multimodal assessment of Parkinson's disease: a deep learning approach*. IEEE journal of biomedical and health informatics 23 (4), 1618-1630.
- Vasquez-Correa, J.C., et al. (2017). *Convolutional Neural Network to Model Articulation Impairments in Patients with Parkinson's Disease*. Interspeech.

A complete list of publications is available on my Google Scholar profile.